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TimeSiam: A Pre-Training Framework for Siamese Time-Series Modeling

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Problem

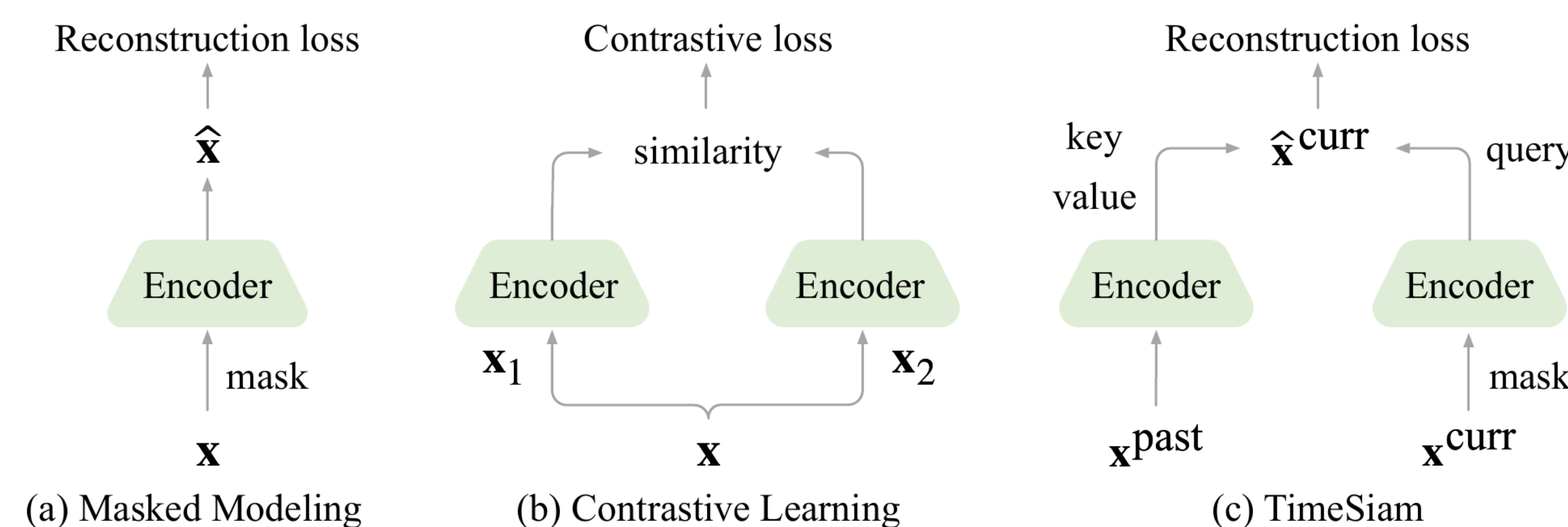
Self-supervised pre-training reuses **masked modeling** and **contrastive learning** from vision and language. On time series both misfire: randomly masking a series or scoring series-wise similarity **distorts or neglects the temporal correlations** intrinsic and crucial to time-series data.

Masking shatters smooth temporal structure; contrastive learning ignores fine-grained within-series correlations and needs augmentations hard to design for time series.

Motivation

- **Unlabeled data is exploding.** IoT sensors and wearables stream ever-growing volumes of time series, ready for label-free pre-training at scale.
- **Prior work wastes the signal.** Each subseries is treated in isolation, overlooking the chronological relationship between temporally distanced observations.

A method that explicitly emphasizes **temporal-correlation modeling** could unlock this data for downstream tasks.



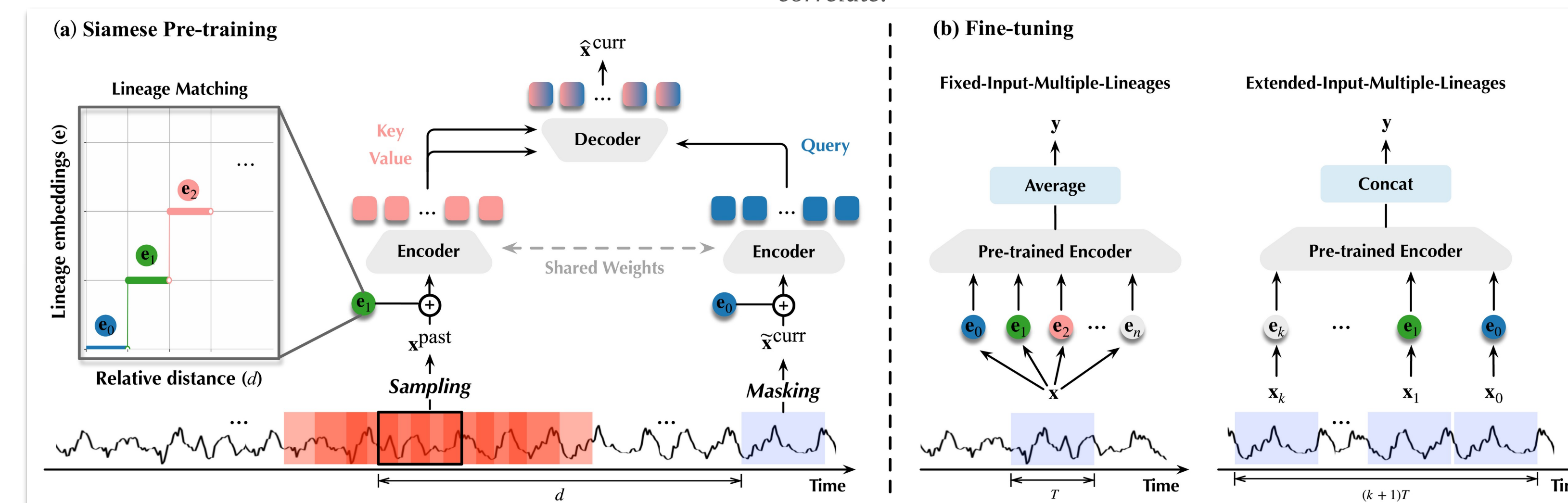
Time-series pre-training paradigms: (a) masked modeling reconstructs a masked series; (b) contrastive learning scores similarity; (c) TimeSiam reconstructs the masked current series from a sampled past observation.

Method

- 1 **Sample** a past and current subseries from the *same* series.
- 2 **Mask** the current subseries with a simple augmentation.
- 3 **Reconstruct**: shared Siamese encoders + a cross-attention decoder rebuild the current series from the past.
- 4 **Lineage embeddings** indexed by past $\rightarrow$ current distance span many temporal gaps.

$$\mathcal{L}_{\text{rec}} = \|\mathbf{x}^{\text{curr}} - \hat{\mathbf{x}}^{\text{curr}}\|_2^2$$

Past-to-current reconstruction is the sole objective — forcing the model to internalize how distant moments correlate.



TimeSiam pre-trains shared Siamese encoders on past/current subseries and fuses learnable lineage embeddings; fine-tuning averages (fixed input) or concatenates (extended input) the lineages.

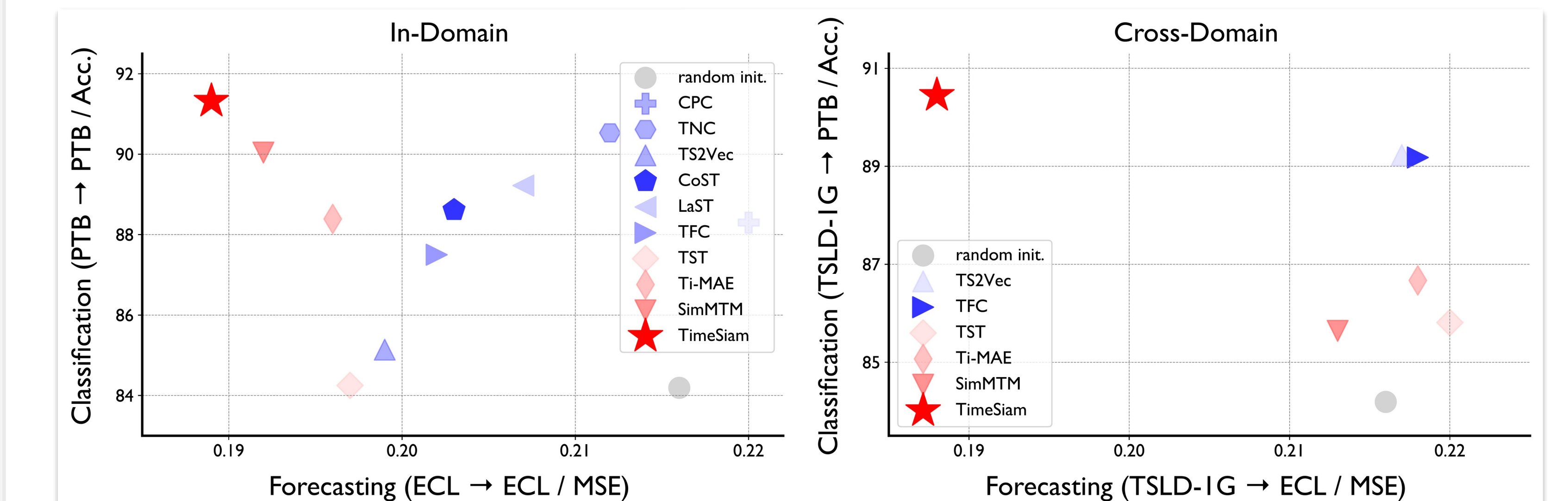
Dataset / Benchmark

13 benchmarks over two tasks, plus two **newly constructed** large-scale multi-domain corpora (TSLD-500M, TSLD-1G with 13.9M examples) for cross-domain pre-training.

ETT×4 Weather Electricity Traffic Exchange AD TDBrain
PTB TSLD-500M TSLD-1G

Key Results

In-domain forecast (avg MSE↓)	Random init.	TimeSiam
ETTh1	0.473	0.429
ETTm1	0.390	0.374
Traffic	0.490	0.453



TimeSiam (red star) beats 8 SSL baselines on forecasting (MSE↓) and classification (Acc↑), in- and cross-domain.

Headline Numbers

+11.5%
average classification-accuracy gain, in-domain

vs random initialization, averaged across the AD, TDBrain and PTB medical classification benchmarks

−5.7% forecast MSE (PatchTST)

13 benchmarks covered

8 SSL baselines beaten

Takeaway

Reframe time-series pre-training as past-to-current reconstruction between Siamese subseries with learnable lineage embeddings — capturing the temporal correlations masking and contrastive methods miss, for simple, general, state-of-the-art transfer across tasks and domains. Backbone-agnostic on iTransformer, PatchTST and TCN.